DESCRIBING LINUX FILE SYSTEM HIERARCHY CONCEPTS

OBJECTIVES

After completing this section, you should be able to describe how Linux organizes files, and the purposes of various directories in the file-system hierarchy.

THE FILE-SYSTEM HIERARCHY

All files on a Linux system are stored on file systems, which are organized into a single *inverted* tree of directories, known as a *file-system hierarchy*. This tree is inverted because the root of the tree is said to be at the *top* of the hierarchy, and the branches of directories and subdirectories stretch *below* the root.

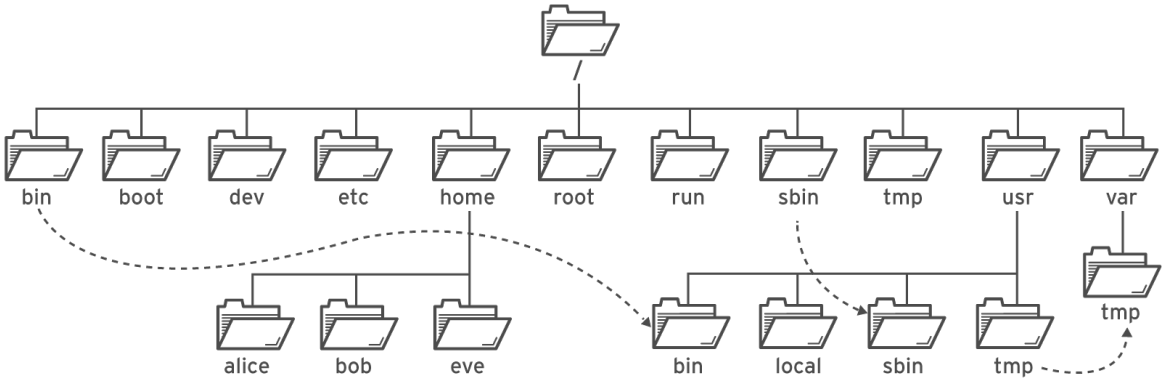

Figure 3.1: Significant file-system directories in Red Hat Enterprise Linux 8

The **/** directory is the root directory at the top of the file-system hierarchy. The **/** character is also used as a *directory separator* in file names. For example, if **`etc`** is a subdirectory of the **/** directory, you could refer to that directory as **`/etc`**. Likewise, if the **`/etc`** directory contained a file named **`issue`**, you could refer to that file as **`/etc/issue`**.

Subdirectories of **/** are used for standardized purposes to organize files by type and purpose. This makes it easier to find files. For example, in the root directory, the subdirectory **`/boot`** is used for storing files needed to boot the system.

NOTE

The following terms help to describe file-system directory contents:

- *static* content remains unchanged until explicitly edited or reconfigured.
- *dynamic* or *variable* content may be modified or appended by active processes.
- *persistent* content remains after a reboot, like configuration settings.
- *runtime* content is process- or system-specific content that is deleted by a reboot.

The following table lists some of the most important directories on the system by name and purpose.

Important Red Hat Enterprise Linux Directories

LOCATION	PURPOSE
`/usr`	Installed software, shared libraries, include files, and read-only program data. Important subdirectories include: • **`/usr/bin`**: User commands. • **`/usr/sbin`**: System administration commands. • **`/usr/local`**: Locally customized software.
`/etc`	Configuration files specific to this system.
`/var`	Variable data specific to this system that should persist between boots. Files that dynamically change, such as databases, cache directories, log files, printer-spooled documents, and website content may be found under **`/var`**.
`/run`	Runtime data for processes started since the last boot. This includes process ID files and lock files, among other things. The contents of this directory are recreated on reboot. This directory consolidates **`/var/run`** and **`/var/lock`** from earlier versions of Red Hat Enterprise Linux.
`/home`	*Home directories* are where regular users store their personal data and configuration files.
`/root`	Home directory for the administrative superuser, `root`.
`/tmp`	A world-writable space for temporary files. Files which have not been accessed, changed, or modified for 10 days are deleted from this directory automatically. Another temporary directory exists, **`/var/tmp`**, in which files that have not been accessed, changed, or modified in more than 30 days are deleted automatically.
`/boot`	Files needed in order to start the boot process.
`/dev`	Contains special *device files* that are used by the system to access hardware.